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Build the matrix B for a three-node bar element with nodal coordinates (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) .

Solution

We can estimate the displacement with the use of shape functions:

$$\mathbf{u}(x, y, z) \approx \mathbf{N}(x, y, z) \mathbf{U}^{(e)} \quad (4.2.2.1)$$

Where $\mathbf{U}^{(e)}$ is a collection of the displacement degrees of freedom.

Analysing a single node in its x displacement, it is true that

$$u_{x(x,y,z)} = \sum_{i=1}^n N_{i(x,y,z)} u_{ix} \quad (4.2.2.2)$$

Deriving in relation to x:

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x} = \sum_{i=1}^n \frac{\partial N_i}{\partial x} u_{ix} \quad (4.2.2.3)$$

Thus, the B matrix represents an element to generalize the relation between the strain vector and displacement matrix:

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \mathbf{B} \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ \vdots \\ u_{nx} \\ u_{ny} \end{bmatrix} \quad (4.2.2.4)$$

which is simply $\varepsilon = \mathbf{B} \mathbf{U}^{(e)}$. B on itself can be represented as a union of matrixes:

$$\mathbf{B} = (\mathbf{B}_1 \ \mathbf{B}_2 \ \mathbf{B}_3 \ \mathbf{B}_4 \ \mathbf{B}_5 \ \dots \ \mathbf{B}_n) \quad (4.2.2.5)$$

Where, for a 2D element

$$\mathbf{B}_i = \begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix} \quad (4.2.2.6)$$

So, for a 3 node element defined in a 2D space, the shape functions are

$$\mathbf{N} = \begin{pmatrix} N_1(\xi) \\ N_2(\xi) \\ N_3(\xi) \end{pmatrix} = \begin{pmatrix} \frac{\xi}{2}(\xi - 1) \\ 1 - \xi^2 \\ \frac{\xi}{2}(\xi + 1) \end{pmatrix} \quad (4.2.2.7)$$

$$(x \ y) = \mathbf{N}^T \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix} \quad (4.2.2.8)$$

Problem is, the shape functions are defined by parametric coordinates (ξ) while the $\mathbf{B}$ matrix derives from the global ones.

Well, it can be shown that

$$\frac{\partial \mathbf{N}}{\partial \mathbf{x}} = \frac{\partial \mathbf{N}}{\partial \xi} \mathbf{J}^{-1} \quad (4.2.2.9)$$